RapidResponse

PRODUCTION PLANNING

Application Deployment Guide

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Kinaxis®

RapidResponse®

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About Kinaxis

Kinaxis is a leading provider of cloud-based subscription software that enables our customers to improve and accelerate analysis and decision-making across their supply chain operations. The supply chain planning and analytics capabilities of our product, RapidResponse, create the foundation for managing multiple, interconnected supply chain management processes. By using the single RapidResponse product instead of combining individual disparate software solutions, our customers gain visibility across their supply chains, can respond quickly to changing conditions, and ultimately realize significant operating efficiencies.

Leaders across multiple industry verticals, including Aerospace & Defense, Automotive, High Tech, Industrial and Life Sciences, rely on RapidResponse for concurrent planning, continuous performance monitoring, and coordinated responses to plan variances across multiple areas of the business.

RapidResponse's configurable service solutions encompass a full spectrum of supply chain related business processes, including such functions as: S&OP, supply planning, capacity planning, demand planning, inventory management, MPS and order fulfillment. As a result, Kinaxis customers have replaced disparate planning and performance management tools and are realizing significant operations performance breakthroughs in planning cycles, supply chain response times and decision accuracy. From a single product, customers are able to easily model varying supply chain conditions to make both long-term and real-time demand and supply balancing decisions quickly, collaboratively, and in line with the shared business objectives of multiple stakeholders.

For more information, visit http://www.kinaxis.com or the Supply Chain Expert community at http://community.kinaxis.com.

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Document Revision History

Version	Changes	Author	Date
1	Updates add to account for new functionality in production planning to support Scheduled SRs and non-reschedulable SRs.	P. Durrani	Feb. 2021
2	Updates to include Firm Orders content as per https://jira.kxsrd.io/browse/RRP-171806	P. Durrani	Dec. 2021

Chapter 1

Introduction

1. Introduction

Production Planning allows you to further control capacity planning by enabling you to now sequence a part through a constraint.

The initial release of production planning in RapidResponse focuses on production (rhythm) wheels to help facilitate a level-loaded production plan. Used in combination with the capacity planning (constraint) application, this new capability enables users to plan parts through a constraint in a predefined part-sequence.

The solution is built on three primary capabilities:

- A production recurrence capability so that production frequency can be easily aligned to a product segmentation strategy.
- A product grouping capability that allows products to be grouped together during production to reduce setup and cleanup activities.
- A product sequencing capability that allows users to predefine the desired order of manufacture to further improve capacity utilization.

This application contains three primary resources, each intended to help users either configure parameters or review results:

- The Production Planning Parameters workbook allows users to create and manage production wheels, product groups, part sequences, and recurrence records.
- The Production Planning Manager workbook provides users with the ability to review results.
- The Production Planning Segmentation workbook provides users with the ability to influence production frequency by assigning products to production segments.

1.1. Objective

The intended audience of this document are solution architects, consultants, and power users. This objective is to provide an overview of the solution and enough detail for you to be able to configure this solution in the field. This document will cover the following topics:

- An overview of new resources needed for configuration and analysis.
- Explanation of the process required to create production wheels, groups, sequences, and recurrence records.
- An overview of the production wheel analytics to help users better understand results.

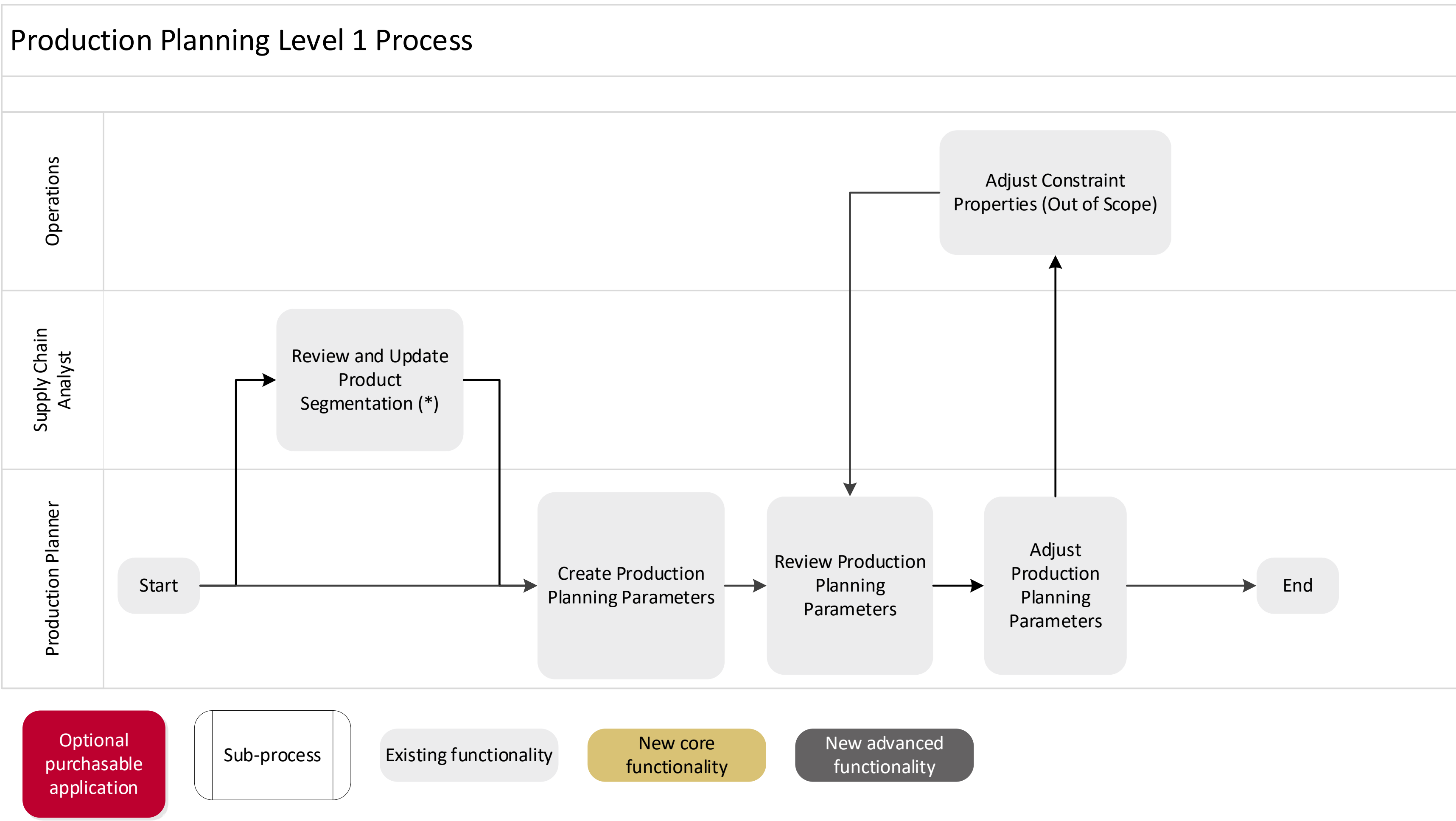
Chapter 2

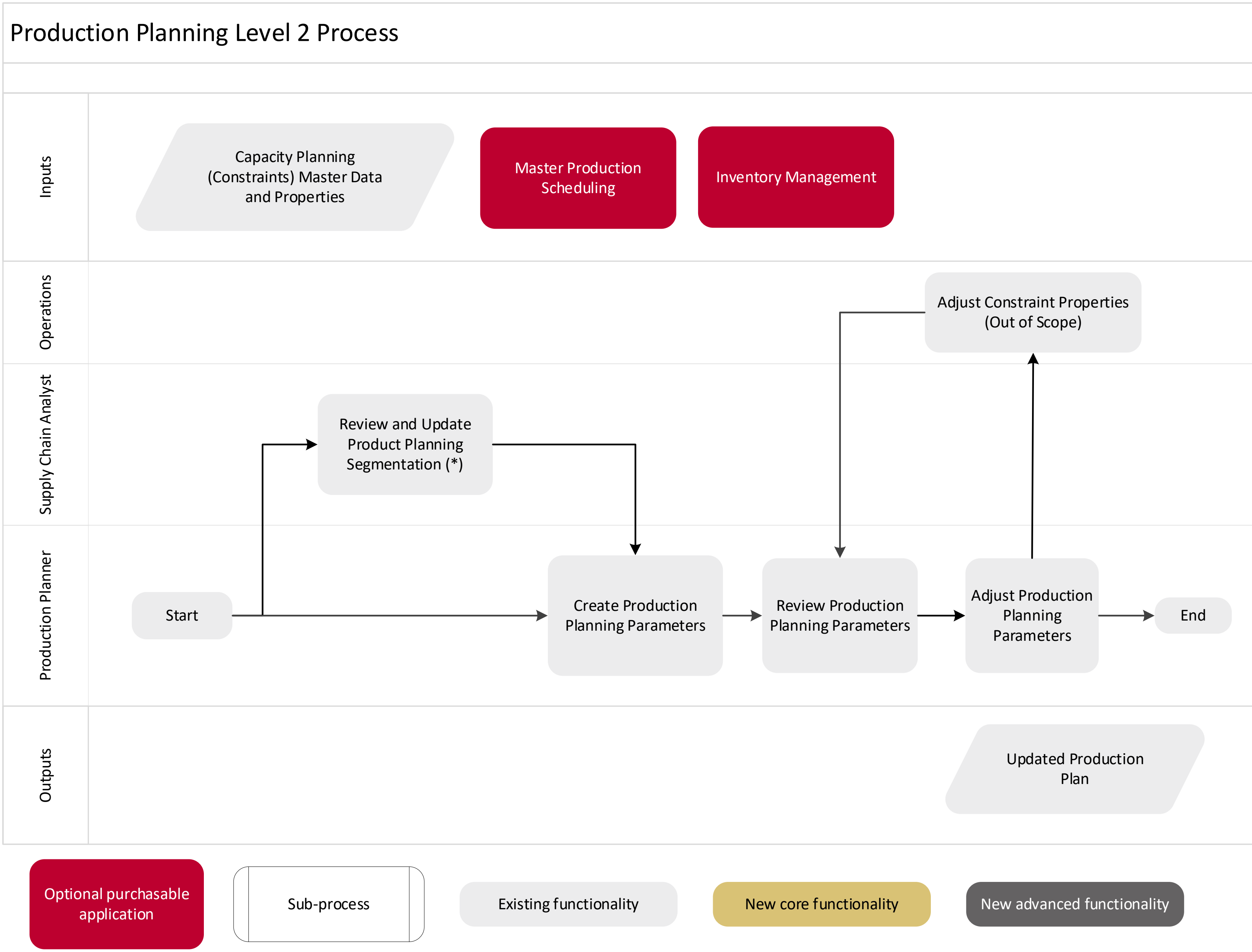
Business Processes

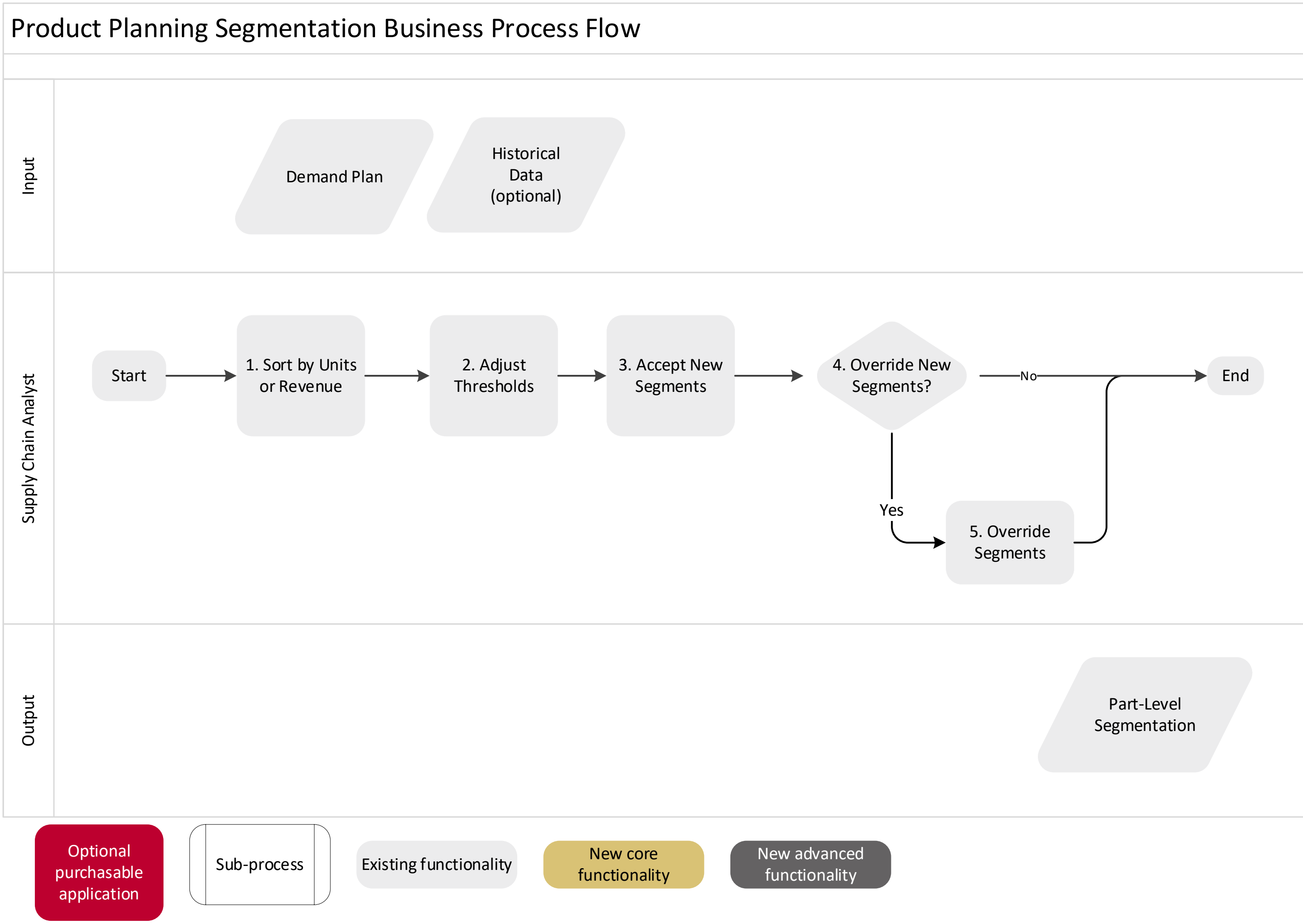
2. Business Process

2.1. Business Process Flow Diagram

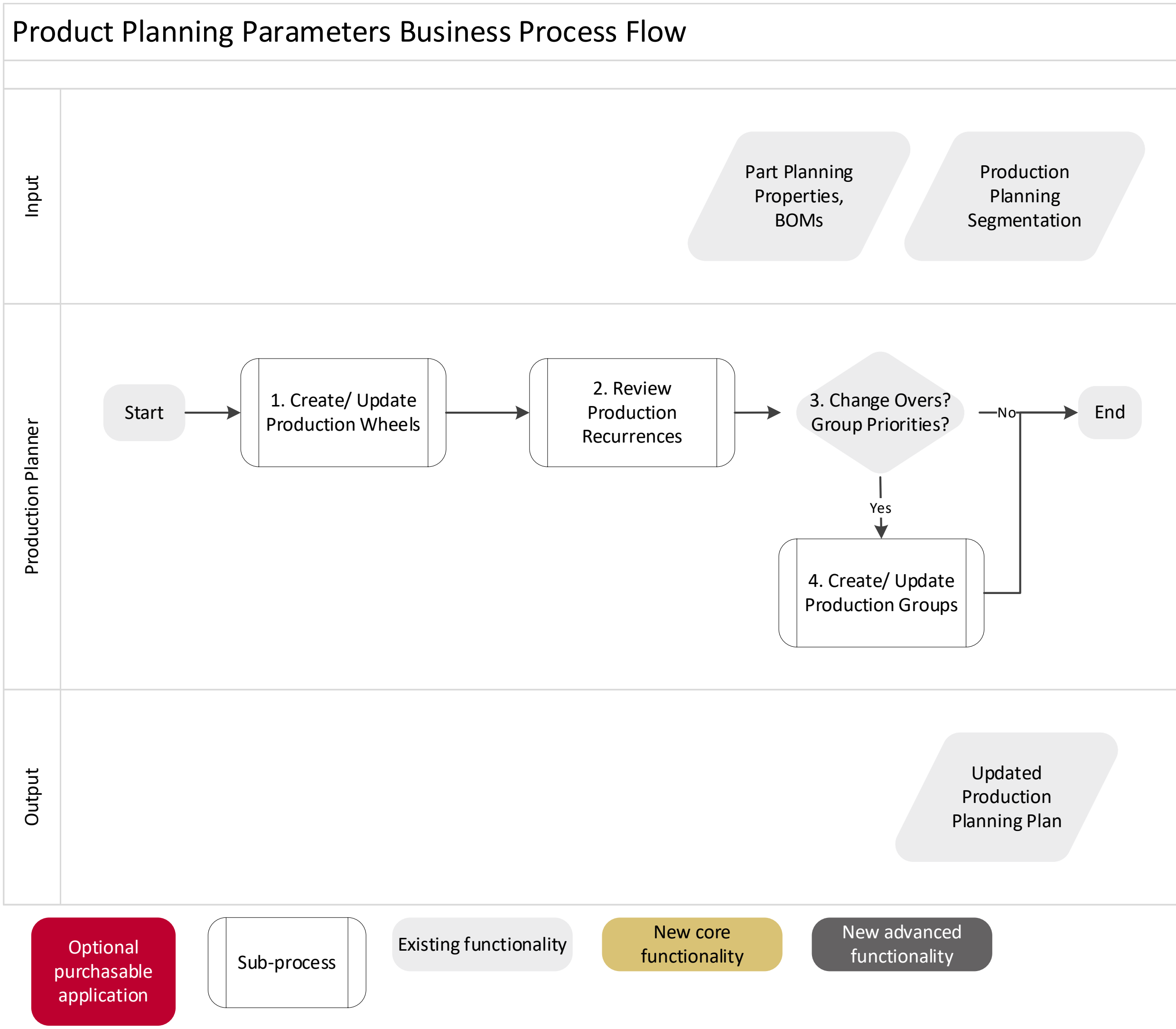
The process flow shown below gives a high level overview of what is contained within the Production Planning application.



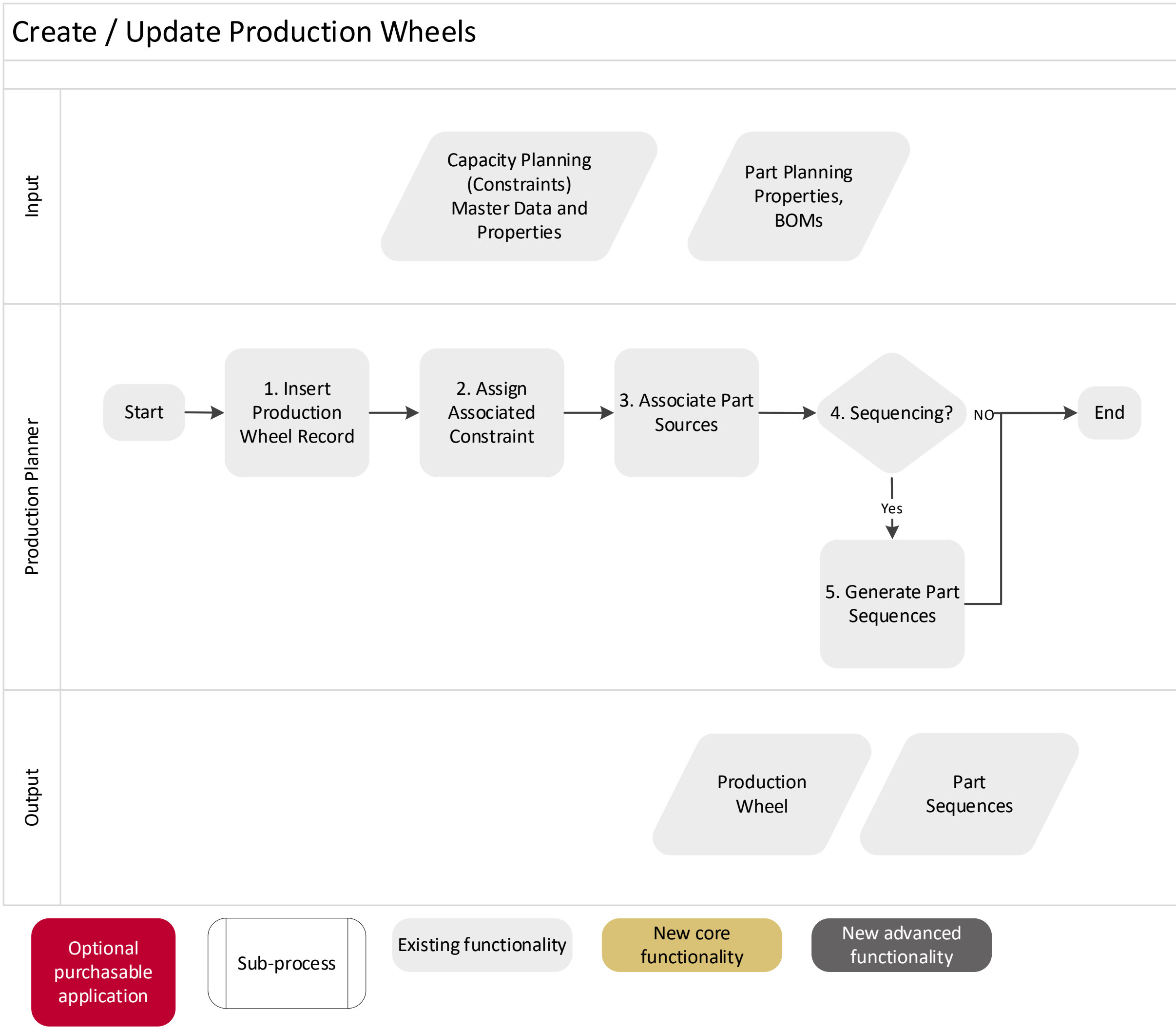

Production Planning Level 2 Process
Inputs
Operations
Supply Chain Analyst
Production Planner
Outputs
Capacity Planning (Constraints) Master Data and Properties
Master Production Scheduling
Inventory Management
Adjust Constraint Properties (Out of Scope)
Review and Update Product Planning Segmentation (*)
Start
Create Production Planning Parameters
Review Production Planning Parameters
Adjust Production Planning Parameters
End
Updated Production Plan
Optional purchasable application
Sub-process
Existing functionality
New core functionality
New advanced functionality



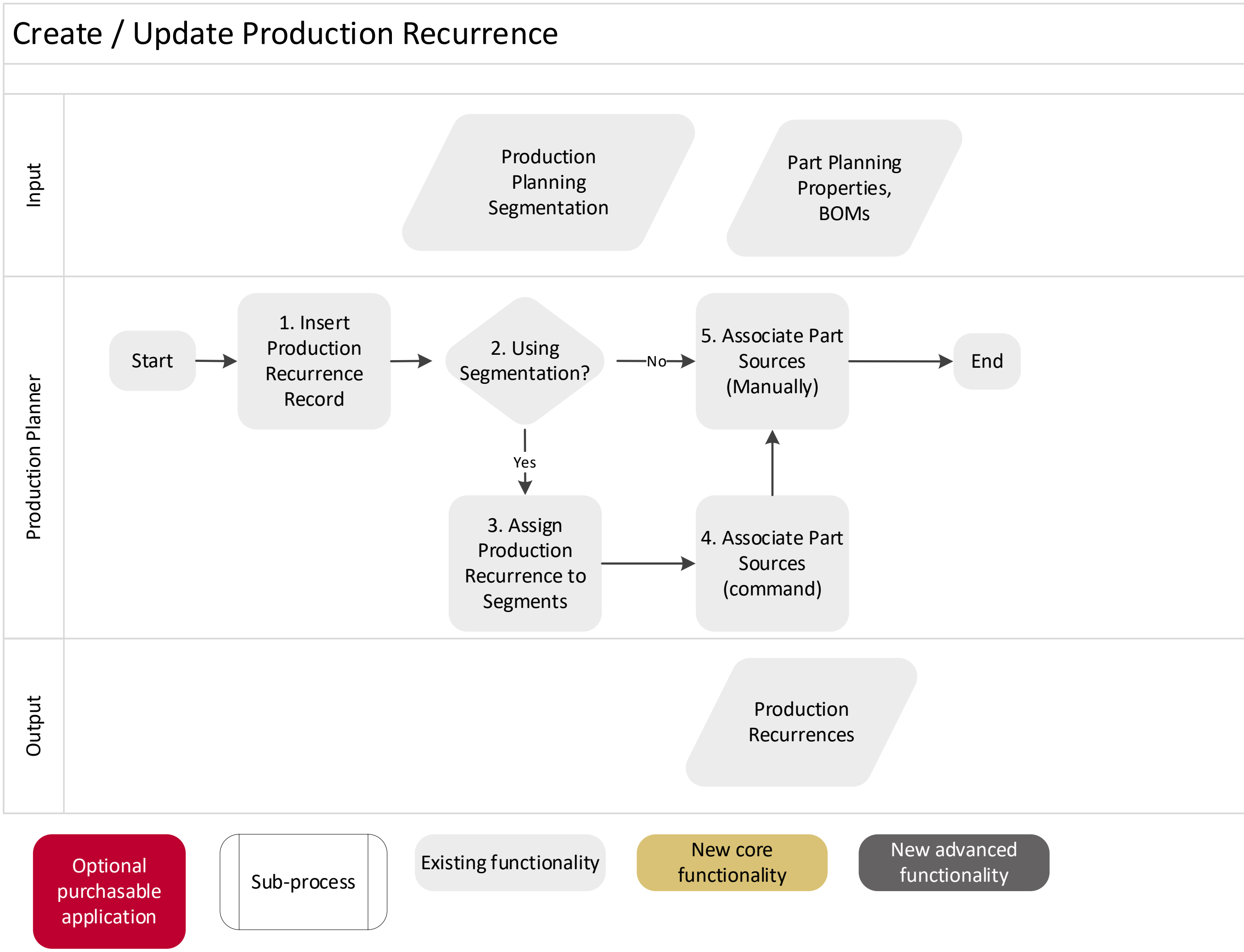
1. **Sort by Units or Revenue:** Users have the option to sort part by their volume or revenue.
2. **Adjust Thresholds:** Using workbook data settings or by drilling on the arrow icons
3. **Accept New Segments:** Using a workbook command
4. **Override New Segments?:** If yes, proceed to the next step, otherwise the flow is complete.
5. **Override Segment:** Drilling into the Override



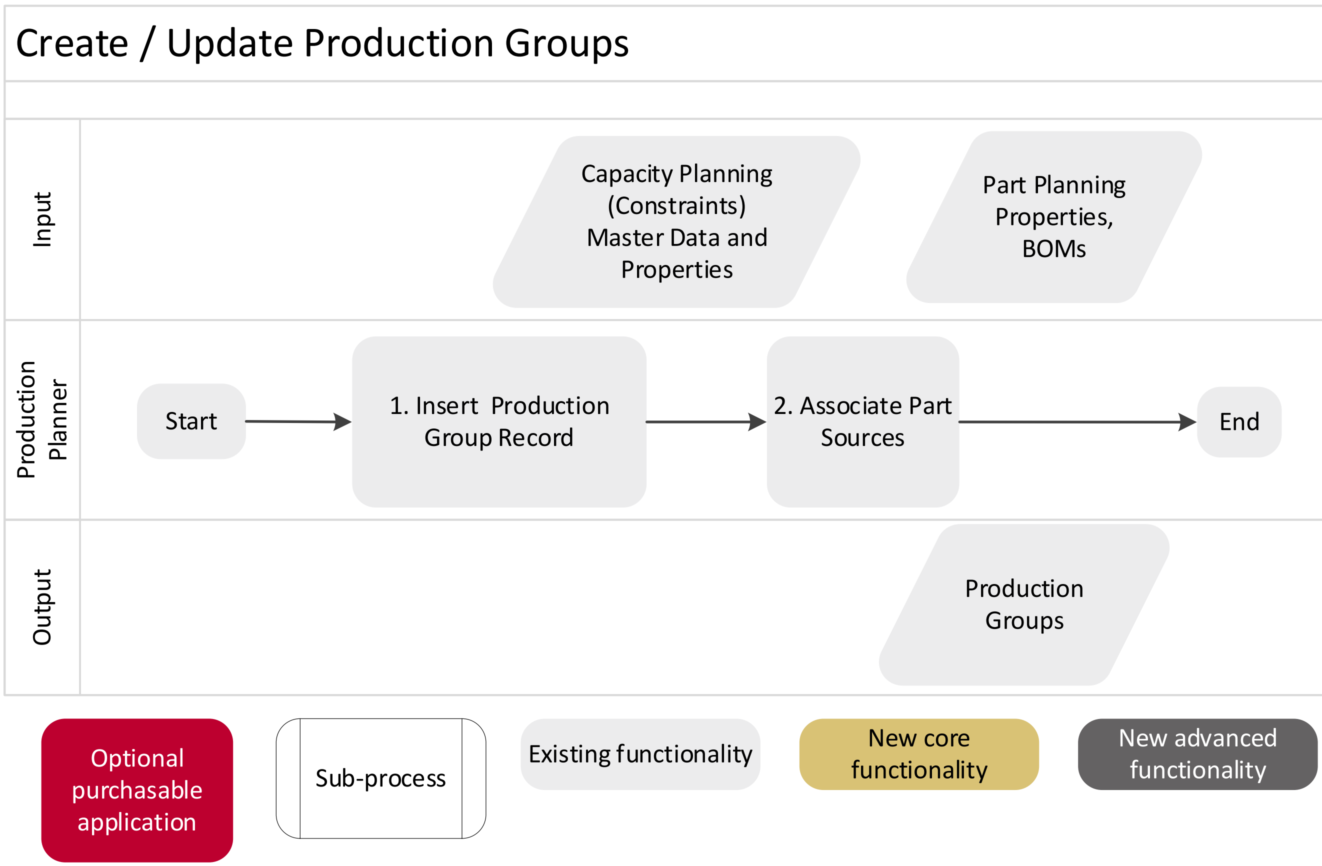
1. **Create / Update Production Wheel:** Using the Production Planning Parameters workbook
2. **Review Production Recurrence**: Using the Production Planning Parameters workbook
3. **Are Changeovers or Group Priorities needed?**: If so proceed to step 4
4. **Create / Update Production Groups:** Using the Production Planning Parameters workbook



1. **Insert production wheel record:** Using a regular insert definition
2. **Assign an associated constraint:** Using the Assign Constraint form from the Production Wheel Worksheets
3. **Associate PartSource:** Drilling to the Part Sources worksheet.
4. **Parts-base Sequencing needed?:** If yes, proceed to step 5
5. **Generate Part Sequences:** Using a command via regular insert definitions on the Part Sequences worksheet.



1. **Insert Production Recurrence Record:** From the Production Recurrence worksheet
2. **Using Segmentation?** If yes, proceed to step 3, otherwise proceed to step 4.
3. **Assign Production recurrence to Segments:** From the Segments worksheet
4. **Associate Part Sources (command):** Running a command from the Segments worksheet
5. **Associate Part Sources (Manually):** Drilling to the Part Sources worksheet.



1. **Insert Production Group Record:** From the Productions Groups worksheet
2. **Associate Part Sources:** Drilling to the Part Sources worksheet.

2.1.1. Inputs

- Capacity Planning (Constraints)
- Part Properties (Sourcing)
- Demand Plan
- Historical Data
- New Input Tables
    - ProductionWheel
    - ProductionWheelType
    - ProductionWheelConstraint
    - ProductionGroup
    - ProductionRecurrence
    - PartSequence
    - Segment
- New input fields added to PartSource
    - ProductionWheel
    - ProductionGroup
    - ProductionRecurrence
- New input field added to PartSolutions
    - Segment

2.1.2. Outputs

- Production Plan (Constraint Usage)
- Changeover (Constraint usage)
- Consumption Sequences (Constraint Usage)
- New Calculated tables
  - ProductionCycle
- New calculated fields added to ConstraintUsed
  - ProductionWheel
  - Type
  - ConsumptionSequence
  - ProductionCycleNumber

Chapter 3

Integration Requirements

3. Integration Requirements

3.1. Data and Integration Requirements

3.1.1. Database Scale

It is imperative that the consultant and customer both understand the database scale that is being created. This typically needs to be a concern when integrating large amounts of historical data or performing heavy disaggregation operations, both are items in the S&OP and Demand Planning Business Services, but that does not mean it should not be thought about and considered when deploying other business service blueprints.

Prior to release 2014.1 RapidResponse supports approximately 1B input records on all input tables in all scenarios. As of release 2014.1 RapidResponse supports approximately 4B input records on all input tables in all scenarios. You do not want to come anywhere near this number for just one table.

For example, forecasting will create a series of detailed ForecastDetail records for eligible PartCustomer records from a very high-level summary perspective. The fact that we are forecasting at a high level does not mean that we are not creating these detailed records and the scope of this can be easily missed. If you are forecasting for 200K parts with an average of 200 customers each, then you are forecasting 40M PartCustomers. Further, if your DP_BaseCalendar is set to weekly and you are forecasting for 2 years, then this will result in 40M * 104 or greater than 4 billion ForecastDetail records. This ***CANNOT*** be supported in RapidResponse and even if it could the performance would be unusable

In addition to the input record limits mentioned above, there is a 4 billion Record ID limit per table across all scenarios that can impact our customers on Historical and ForecastDetail tables. An input record is assigned a record ID at creation that is simply a counter for each table. The Record ID values are not reused and (except in one very special case) they are not recovered by a server restart or data update operation. So, if you were to import or create 600M input records on a table (like ForecastDetails or, more often, HistoricalDemandSeriesDetail or HistoricalDemandActuals), and then delete them and bring in another set on the next cycle, you would never hit the 4 billion records at any given time. However, you would still be consuming the record IDs 600M at a time. Therefore, on your 7th update cycle, you would crash the RapidResponse server, having exhausted all available Record IDs.

3.1.2. Detailed Data Requirements

3.1.2.1. ProductionWheel

- There has to be at least one record for this table.
- EffectiveInDate cannot be past, future or undefined, an actual date needs to be provided. Periodic maintenance will be needed to roll this date as necessary.
- CalendarInterval and Calendar define the **cycle** length (ex: ‘1’ ‘week’)
- When there are multiple wheels in the same time frame, priority determines the order in which the wheels are processed. If the priorities are equal, Name is used.

3.1.2.2. ProductionWheelConstraint

- There has to be one record per wheel. The constraint needs to be a **constrained** constraint.
- MinimumConstraintConsumption represents a fraction of constraint available per cycle. This value is used in conjunction with ProductionWheelPolicy.UnusedConstraintRule. After processing supplies that originally belong to a cycle, if the amount of constraint used falls below the MinimumConstraintConsumption, this application will try to increase constraint consumption (following UnusedConstraintRule). Note that absolute preplan limit and Preplan Limit are still respected.

3.1.2.3. ProductionGroup

- This table is required if changeover are to be used. It is not mandatory and other functionality of the production wheel can be used without defined groups.
- MaximumConstraintConsumption represents a fraction of constraint available per cycle. This setting is used in conjunction with ProductionWheelPolicy.UnsatisfiedDemandsRule to limit how much constraint can be used by a Production Group in any given cycle. This could be used to prevent a group from hogging a constraint. Note, if this setting comes into conflict with PartSource lot sizes, we will choose to meet PS lot sizing while staying as close to this value as possible.

3.1.2.4. ProductionRecurrence

- This table is required to control the recurrence of production across cycles. It is not mandatory and other functionality of the production wheel can be used without defined recurrences.
- Offset are measured from the first cycle. An offset of 1 means the first cycle is to be skipped. A cycle of 3 means, 3 cycles are to be skipped.
- Number of Recurrence of 1 means to build every cycle. Recurrence of 2 means every other cycle.

3.1.2.5. PartSequence

- The order this part is processed in the production wheel. Smaller numbers are processed first.
- If two parts share the same sequence number for a given production wheel, sequence will be defined by other applicable priorities (Production group, other RapidResponse policies and supply sorting criteria).

3.1.2.6. Scheduled Receipts

- Scheduled SRs should only be used in a production wheel when needed. For example, if a planner has to override production for a situation such as a regulatory inspection.
- Scheduled SRs and non-reschedulable SRs should not be placed in the same production cycle. Modify the due date for the non-reschedulable SR to move it out of a cycle that includes a scheduled SR.
- For scheduled SRs, always define a date for **StartDate**. If left undefined, the value calculated by RapidResponse might produce unexpected results.

3.1.3. Control Table Configuration

3.1.3.1. Production Wheel Type

- This table defines the rules and values used to manage production wheels.
- A new worksheet has been added to the Control Tables and Control Sets workbooks called Production Wheel Type
- Currently the rules than can be set are:
  - **UnsatisfiedSupplyRule:** This rule controls how to handle supplies for which we don't have enough capacity within a cycle
    - **Defer:** Unsatisfied supplies are handled in the next cycle where the part is eligible to be built. This the default rule
    - **Ignore:** Unsatisfied quantities are ignored and will not be built. The demands will have an available date of Future.
  - **UnusedConstraintRule:** This rules controls how to allocate surplus capacity within a cycle.
    - **SatisfyFutureSupply:** Satisfy supplies that would normally be produced in future cycles for parts that are eligible to be built in the current cycle of this wheel. This the default rule
    - **Ignore:** Ignore the value set for Minimum Constraint Consumption. Surplus capacity is left unused.

3.1.3.2. SupplyStatus

- For nonschedulable SRs, set **AvailableDateLimit** to **CalcRequestStart** to minimize gaps in the middle of cycles.
- For scheduled SRs, set **IgnoreStartDate** to **N** to have RapidResponse use **ScheduledReceipt.StartDate**. The **StartDate** should have values defined.

3.1.4. Configuration Points

3.1.4.1. User Set Up

- Out of the Box resources are shared with users in the Production Planning group

3.2. Data Integrity Requirements

3.2.1. Production Wheel

- EffectiveInDate cannot be past, future, or undefined.
- Avoid overloading individual cycles with scheduled SRs (if using).

3.2.2. PartSource

- Only partsources with Type.PlannedOrderSupplyType.Source = Make are relevant for production planning .
- The PartSource effectivity window must overlap with the ProductionWheel effectivity window.
- At least one source constraint per part source is needed.
- The SourceConstraint record should match the constraint associated with the production wheel.
- The SourceConstraint effectivity window must overlap with the ProductionWheel effectivity window.
- The PartSource reference for ScheduledReceipt needs to have Production Wheel NOT = null and have a reference to a valid ProductionWheel record.

3.2.3. Production Wheel Constraint

- Constraint needs to be have a constrained processing rule.
- Constraint should have the same calendar as the production wheel.
- MinimumConstraintConsumption should be a value between 0 and 1.

3.2.4. Production Groups

- MaximumConstraintConsumption should be a value between 0 and 1.
- Negative changeovers values will result in zero changeover.

3.2.5. Production Recurrence

- Offset values should be greater than 0.
- Recurrence values should be greater than 1 .

3.2.6. SourceConstraint

- EffectiveInDate should not fall in between cycles.

Chapter 4

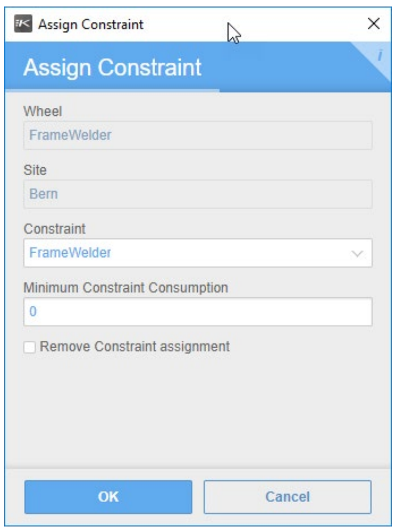
Elaborated Resources

4. Elaborated Resources

4.1. Forms

Outline the standard Forms provided with the application, and detail any special considerations. Insert the standard Form snippets(s) in this section, if applicable. If additional configuration details are required, include those as regular text. If no Forms are provided with the application, identify this section as Not Applicable.

4.1.1. Assign Constraint



This form is used to insert/edit/delete records in the ProductionWheelConstraint table. It's accessible from the Production Wheels worksheet in the Production Planning Parameters workbook

4.2. Scripts

4.2.1. Assign Constraint

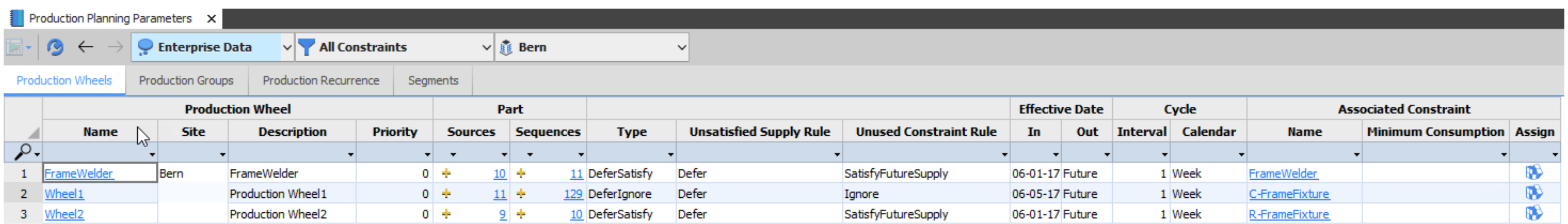
This script supports the Assign Constraint form.

4.3. Workbooks

4.3.1. Production Planning Parameters

Workbook Name	Production Planning Parameters
Purpose	Use this workbook to view and set up parameters for production wheels, production groups, production recurrences, and the segments assigned to parts used in the wheels.
Is Standard PDR?	Yes

4.3.1.1. Production Wheels

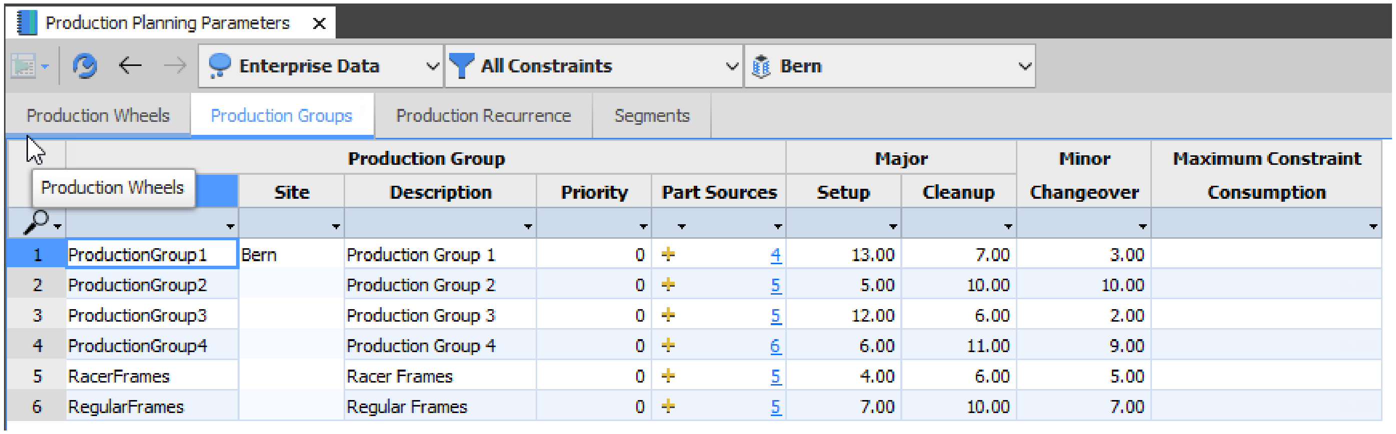


	Production Wheel				Part					Effective Date		Cycle		Associated Constraint		
	Name	Site	Description	Priority	Sources	Sequences	Type	Unsatisfied Supply Rule	Unused Constraint Rule	In	Out	Interval	Calendar	Name	Minimum Consumption	Assign
1	FrameWelder	Bern	FrameWelder	0	10	11	DeferSatisfy	Defer	SatisfyFutureSupply	06-01-17	Future	1	Week	FrameWelder		
2	Wheel1		Production Wheel1	0	11	129	DeferIgnore	Defer	Ignore	06-05-17	Future	1	Week	C-FrameFixture		
3	Wheel2		Production Wheel2	0	9	10	DeferSatisfy	Defer	SatisfyFutureSupply	06-01-17	Future	1	Week	R-FrameFixture		

This worksheet displays details on the parameters used to set up production wheels.

- PartSources available will be affected by the Constraint filter, as only filtered source constraint records are tallied.
- EffectiveInDate cannot be undefined, past or future. Currently there is no visual warnings to the user about this.
- Drilling from the number of either part sources or partsequences will filter (qbe) to the selected parts. Drilling from the yellow plus sign will drill without applying qbe filters.

4.3.1.2. **Production Groups**

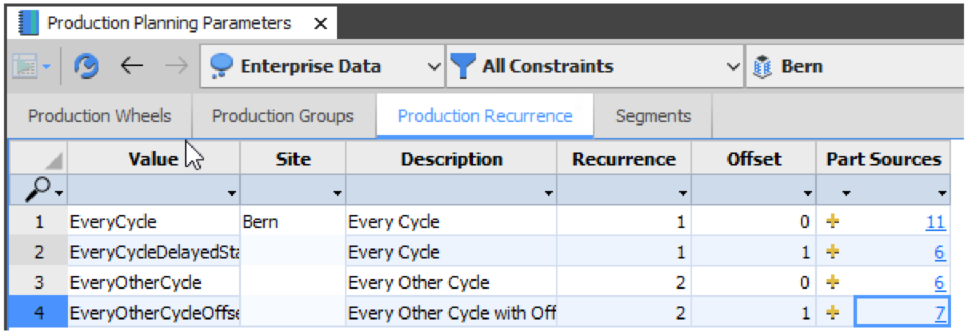


	Production Group	Site	Description	Priority	Part Sources	Major Setup	Major Cleanup	Minor Changeover	Maximum Constraint Consumption
1	ProductionGroup1	Bern	Production Group 1	0	4	13.00	7.00	3.00	
2	ProductionGroup2		Production Group 2	0	5	5.00	10.00	10.00	
3	ProductionGroup3		Production Group 3	0	5	12.00	6.00	2.00	
4	ProductionGroup4		Production Group 4	0	6	6.00	11.00	9.00	
5	RacerFrames		Racer Frames	0	5	4.00	6.00	5.00	
6	RegularFrames		Regular Frames	0	5	7.00	10.00	7.00	

This worksheet displays details on the parameters used to set up the production groups used in a production wheel. Use this worksheet to add, edit, or remove the production groups available to assign to a production wheel.

- PartSources available will be affected by the Constraint filter, as only filtered source constraint records are tallied.
- Drilling from the number of part sources or partsequences will filter (qbe) to the selected parts. Drilling from the yellow plus sign will drill without applying qbe filters.
- Setup and Cleanup applies between production groups, minor changeover applies in between parts of the same group
- At the moment, there are no sequent dependent changeovers

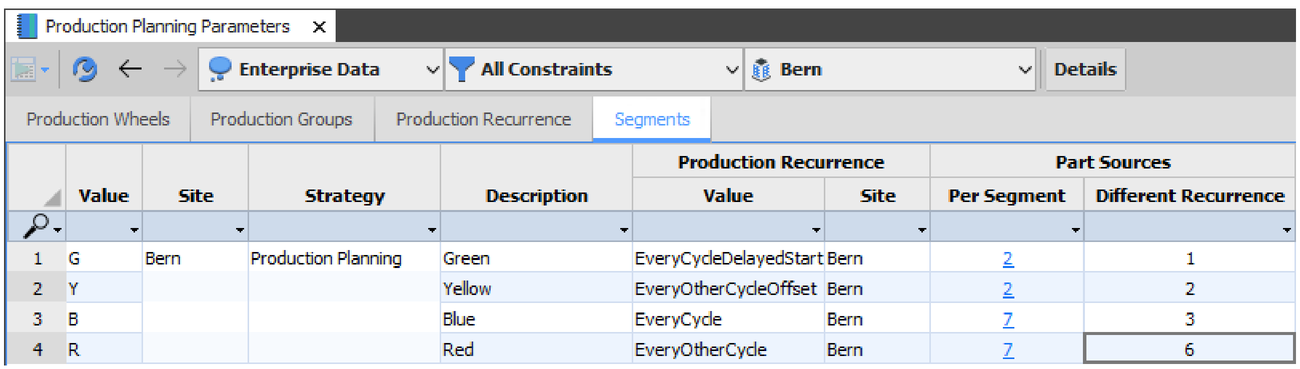
4.3.1.3. Production Recurrence



	Value	Site	Description	Recurrence	Offset	Part Sources
1	EveryCycle	Bern	Every Cycle	1	0	11
2	EveryCycleDelayedSt		Every Cycle	1	1	6
3	EveryOtherCycle		Every Other Cycle	2	0	6
4	EveryOtherCycleOffs		Every Other Cycle with Off	2	1	7

This worksheet displays details on the parameters that define the production recurrences available in your instance of RapidResponse. Use this worksheet to review and add, edit, or remove the production recurrences assigned to part segments.

4.3.1.4. Segments



					Production Recurrence		Part Sources	
	Value	Site	Strategy	Description	Value	Site	Per Segment	Different Recurrence
1	G	Bern	Production Planning	Green	EveryCycleDelayedStart	Bern	2	1
2	Y			Yellow	EveryOtherCycleOffset	Bern	2	2
3	B			Blue	EveryCycle	Bern	7	3
4	R			Red	EveryOtherCycle	Bern	7	6

This worksheet displays details for the part sources associated with a production wheel, production group, production recurrence, or segment.

4.3.1.5. **Part Sources**

Part Sources

	Part		Production Wheel		Production Group		Production Recurrence	
	Name	Site	Name	Site	Name	Site	Value	Site
1	BIKES	Bern	FrameWelder	Bern	ProductionGroup1	Bern	EveryCycle	Bern
2	BRAKE-01	Bern	Wheel2	Bern	ProductionGroup2	Bern	EveryCycle	Bern
3	C-FRAME-STD	Bern	Wheel1	Bern	ProductionGroup3	Bern	EveryCycle	Bern
4	CRUISER	Bern	FrameWelder	Bern	ProductionGroup4	Bern	EveryOtherCycle	Bern
5	DR-0601	Bern	Wheel1	Bern	ProductionGroup1	Bern	EveryCycleDelayedSt	Bern
6	DR-0701	Bern	Wheel2	Bern	ProductionGroup2	Bern	EveryOtherCycleOffs	Bern
7	DR-0801	Bern	FrameWelder	Bern	ProductionGroup3	Bern	EveryCycle	Bern
8	DR-0802	Bern	Wheel1	Bern	ProductionGroup4	Bern	EveryCycleDelayedSt	Bern

This worksheet allows the editing of Production Planning related fields in the PartSource table (Production Wheel, Production Group and Production Recurrence). Inserting (creating a new partsource) and deleting records is disabled.

Note, in order to enter a Name/Value the associated site columns needs to be already populated with a site where there are names/values defined.

4.3.1.6. **Part Sequences**

Part Sequences

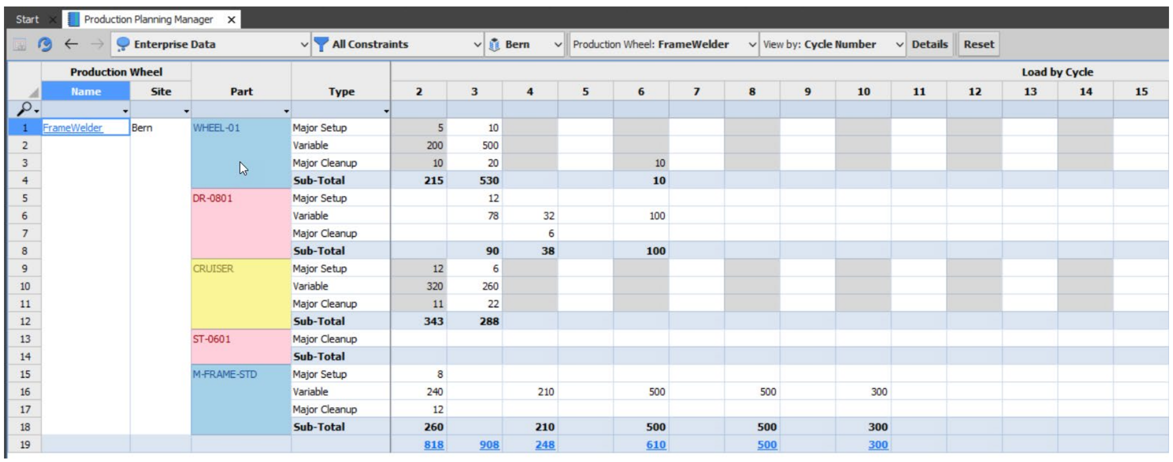
	Production Wheel		Part	
	Name	Site	Name	Sequence
	FrameWelder	Bern		
1	FrameWelder	Bern	WHEEL-01	
2			BIKES	1
3			BR-0101	2
4			CRUISER	3
5			DR-0801	4
6			DRIVE-01	5
7			FR-0101C	6
8			M-FRAME-STD	7
9			R-FRAME-STD	8
10			ST-0601	9
11			STEER-02	10

This worksheet allows for the creation/review of Part Sequences. If the command to auto populate the sequences is run, records will be created with sequence equal to zero. The column formatting will turn the background yellow to indicate this, therefore is suggested to use sequence number different than zero.

4.3.2. Production Planning Manager

Workbook Name	Production Planning Manager
Purpose	Identify possible planning issues involving production wheels.
Is Standard PDR?	Yes

4.3.2.1. Constraint Plan



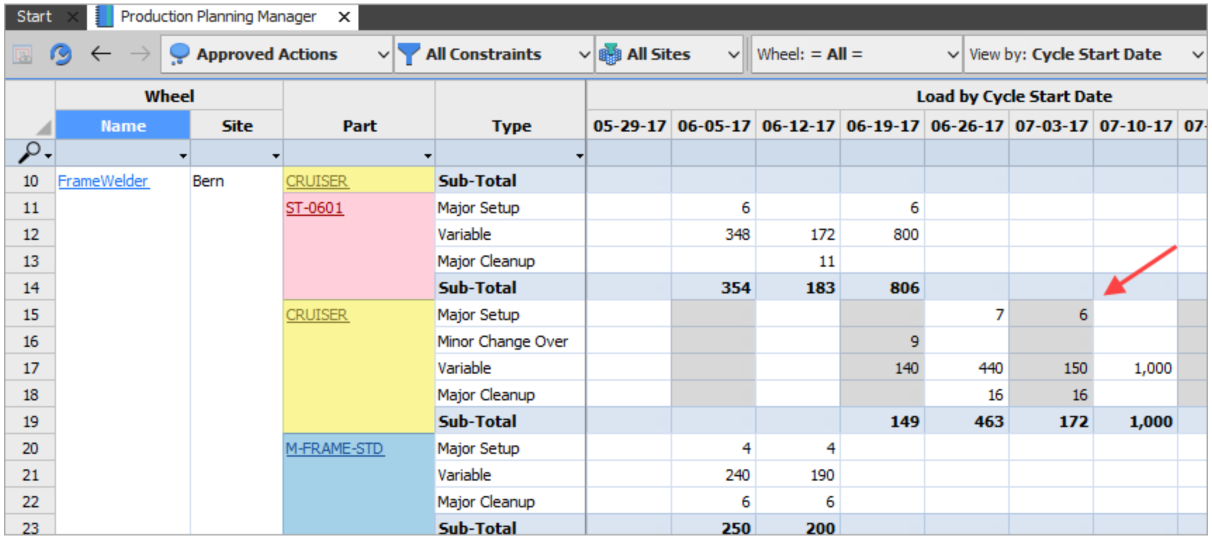
	Production Wheel Name	Site	Part	Type	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	FrameWelder	Bern	WHEEL-01	Major Setup	5	10												
2				Variable	200	500												
3				Major Cleanup	10	20			10									
4				**Sub-Total**	**215**	**530**			**10**									
5			DR-0801	Major Setup		12												
6				Variable		78	32		100									
7				Major Cleanup			6											
8				**Sub-Total**		**90**	**38**		**100**									
9			CRUISER	Major Setup	12	6												
10				Variable	320	260												
11				Major Cleanup	11	22												
12				**Sub-Total**	**343**	**288**												
13			ST-0601	Major Cleanup														
14				**Sub-Total**														
15			M-FRAME-STD	Major Setup	8													
16				Variable	240		210		500		500		300					
17				Major Cleanup	12													
18				**Sub-Total**	**260**		**210**		**500**		**500**		**300**					
19					818	908	248		610		500		300					

Depending on the value of View By, user can see the Constraint Plan by

- Cycle Start Date
- Cycle Number
- Load Date

Load Date uses the advanced bucket settings, so it will enable the bucket settings button on the toolbar.

Any breaks in recurrence are identified by constraint values highlighted in gray.



There a few drills from this worksheets:

- Constraint Plan by Load date:
    - Constraint Load by Date: Drill from Wheel Name column. A summary chart listing Productive (Variable + Fixed) vs Changeover loads by date bucket
    - Supply Summary: Drill from Part. A summary view of Demand, Supply & Inventory by date bucket
    - Load Details: Drill from the Totals Load (bottom row by Wheel)
- Constrain Plan by Cycle Start Date:
    - Constraint Utilization by Part (Cycle Start Date): Drill from Wheel. A chart showing constraint utilization by cycle start date. It shows the sequence of consumption. This chart gets busy quickly depending the number of parts and changeovers
    - Load Details: Drill from the Totals Load (bottom row by Wheel)
- Constrain Plan by Cycle:
    - Constraint Utilization by Part (Cycle): Drill from Wheel. A chart showing constraint utilization by cycle. It shows the sequence of consumption. This chart gets busy quickly depending the number of parts and changeovers
    - Load Details: Drill from the Totals Load (bottom row by Wheel)

4.3.2.2. **Load Details**

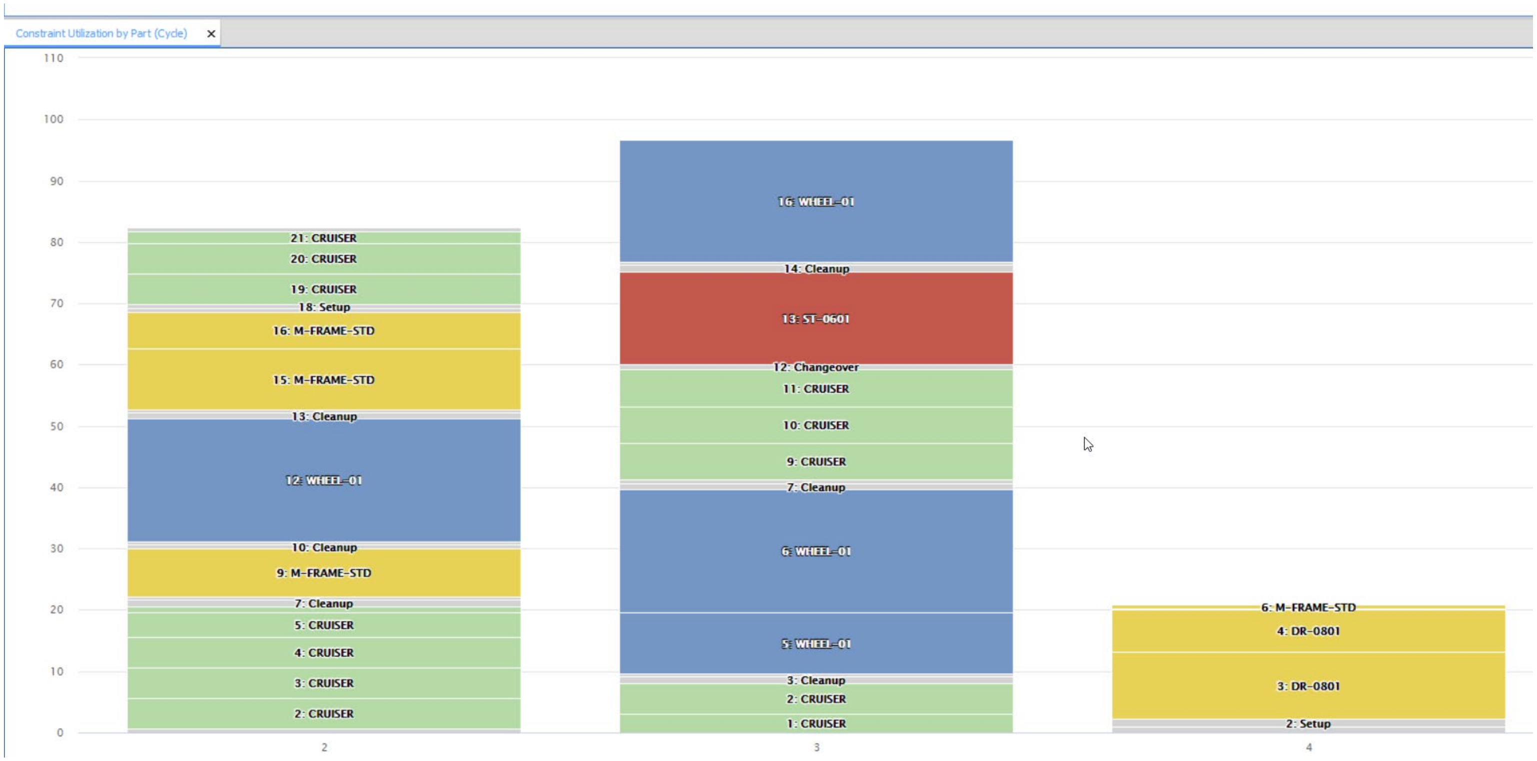
Load Detail

	Production Wheel		Cycle		Constraint			Production Group	Part	Load				
	Name	Site	Number	Start	Name	Site	Available	Name	Name	Date	Type	Sequence	Load	Utilization
	FrameWelde		6											
1	FrameWelder	Bern	6	07-03-17	FrameWelder	Bern	1,000	ProductionGroup3	DR-0801	07-03-17	Variable	1	100.00	10.0%
2								RacerFrames	M-FRAME-STD	07-03-17	Variable	2	28.00	2.8%
3												3	72.00	7.2%
4										07-04-17	Variable	4	100.00	10.0%
5										07-05-17	Variable	5	100.00	10.0%
6										07-06-17	Variable	6	80.00	8.0%
7												7	20.00	2.0%
8										07-07-17	Variable	8	100.00	10.0%
9								ProductionGroup2	WHEEL-01	07-04-17	Major Cleanup	9	10.00	1.0%
10			Sum										610.00	61.0%
11	Sum												610.00	

This worksheet reports details about the constraint usage. If Utilization values result in a sum value over 100%, then the wheel has consumed more than the allocated constraint because of one or more large scheduled SRs are included in the wheel.

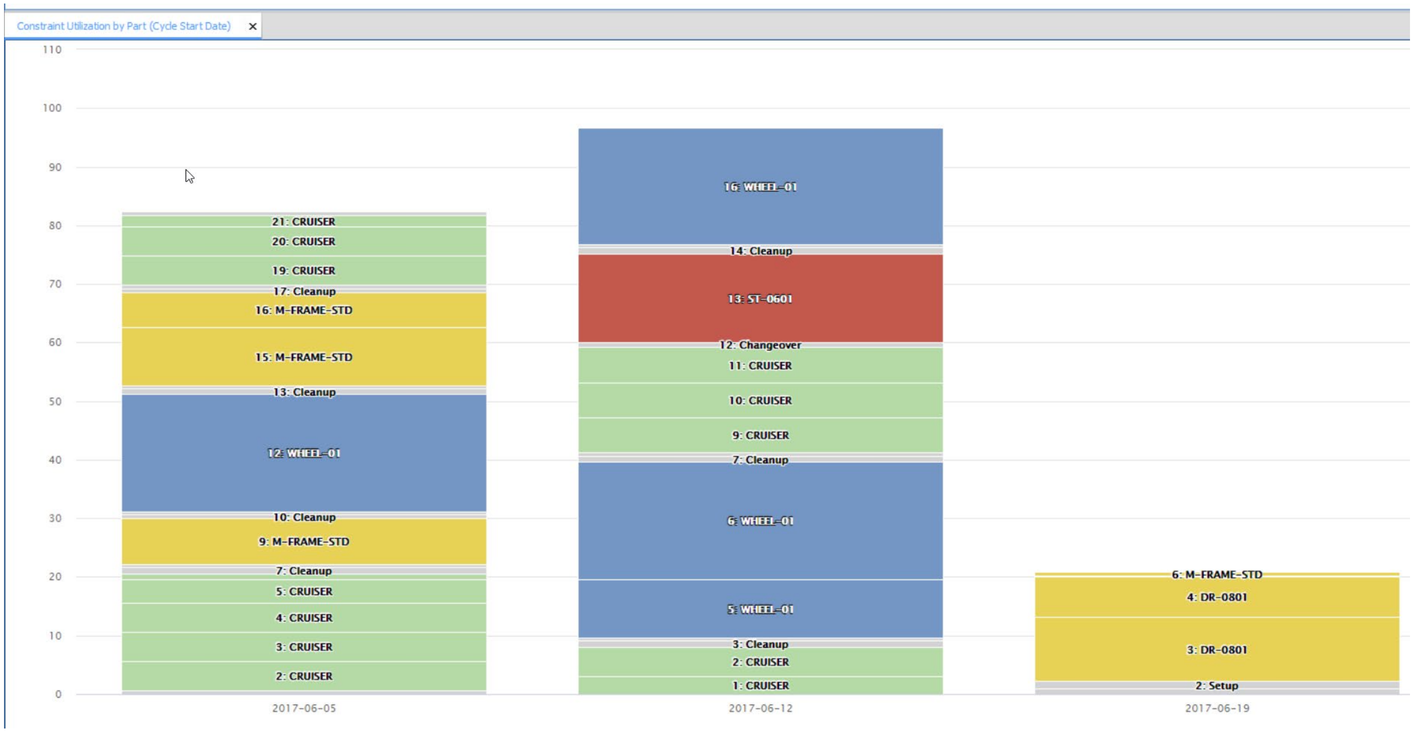
For a given Production Wheel, records are sorted by Cycle and Load Sequence. More details (including Supply Order info) can be revealed by toggling the Details button.

4.3.2.3. **Constraint Utilization by Part (Cycle)**



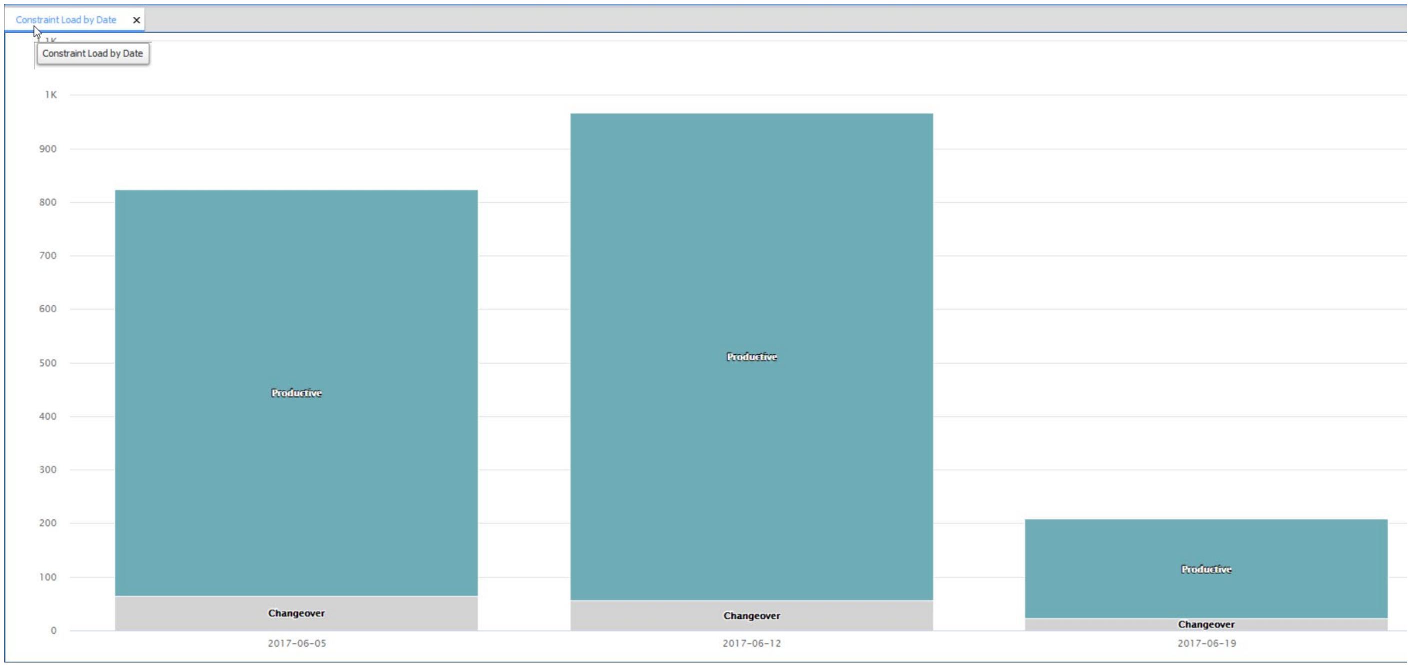
This chart reports constraint utilization by cycle. Parts are stacked according to their consumption sequence within the cycle. The x axis reports Cycle Number, while the Y axis reports Utilization (%). No data will appear if all production wheel are selected (“= All =” value is selected for the Production Wheel setting)

4.3.2.4. Constraint Utilization by Part (Cycle Start Date)



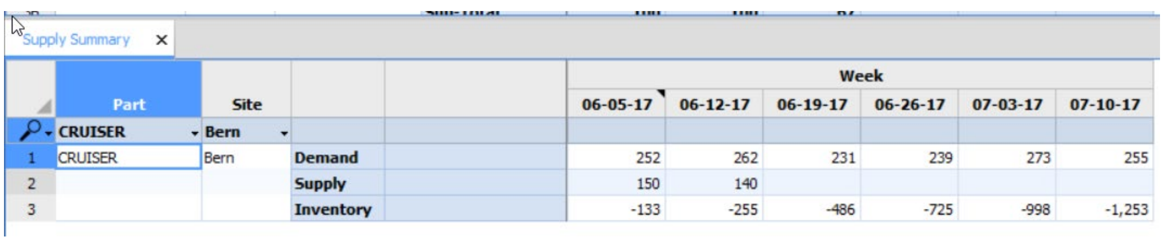
This chart reports constraint utilization by cycle start date. Parts are stacked according to their consumption sequence within the cycle. The x axis reports Cycle Start Date, while the Y axis reports Utilization (%). No data will appear if all production wheel are selected ("= All =" value is selected for the Production Wheel setting)

4.3.2.5. Constraint Load by Date



This chart reports constraint load. The x axis reports date buckets dates, while the Y axis reports constraint load (in units of constraint). No data will appear if all production wheel are selected ("= All = "value is selected for the Production Wheel setting)

4.3.2.6. Supply Summary


Supply Summary

	Part	Site			Week 06-05-17	06-12-17	06-19-17	06-26-17	07-03-17	07-10-17
	CRUISER	Bern								
1	CRUISER	Bern	Demand		252	262	231	239	273	255
2			Supply		150	140				
3			Inventory		-133	-255	-486	-725	-998	-1,253

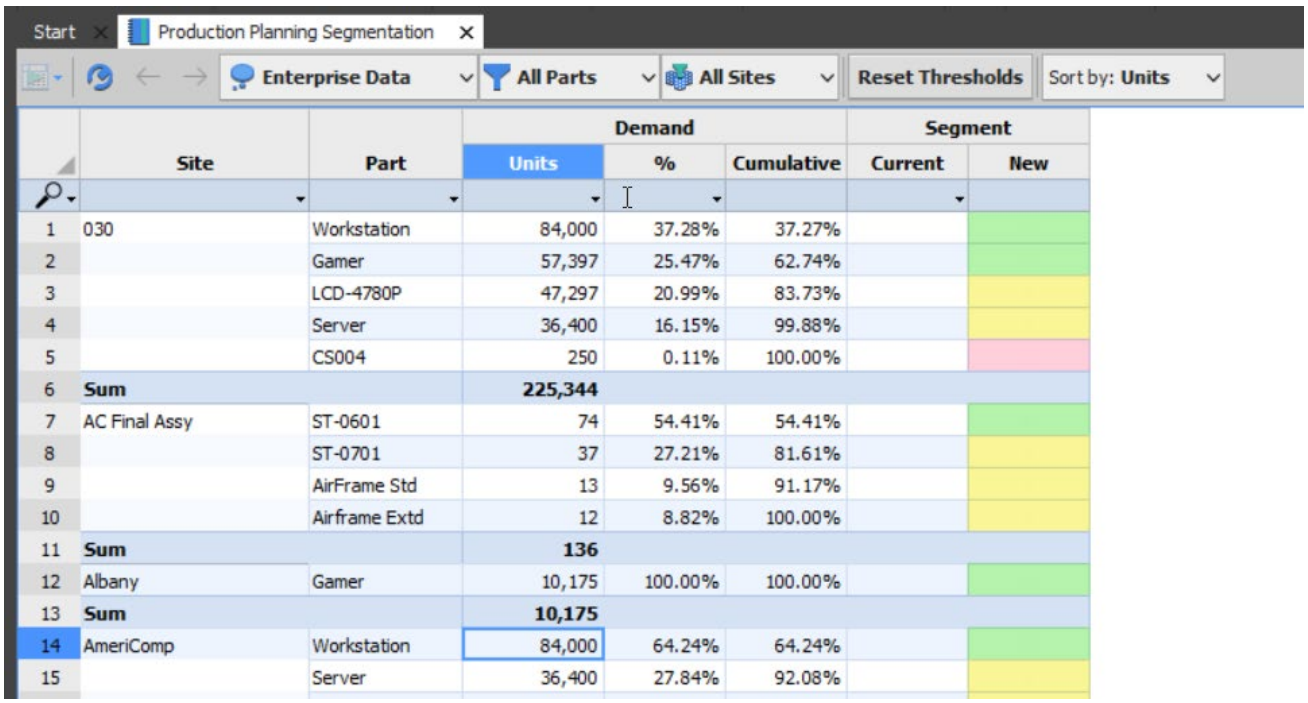
This worksheet shows a breakdown of the demand, supply and inventory by part by date.

4.3.3. Production Planning Segmentation

Workbook Name	Production Planning Segmentation
Purpose	Analyze and group parts into segments
Is Standard PDR?	Yes

Production Planning

Sorted by units - Multiple Sites selected



	Site	Part	Demand Units	Demand %	Demand Cumulative	Segment Current	Segment New
1	030	Workstation	84,000	37.28%	37.27%		
2		Gamer	57,397	25.47%	62.74%		
3		LCD-4780P	47,297	20.99%	83.73%		
4		Server	36,400	16.15%	99.88%		
5		CS004	250	0.11%	100.00%		
6	Sum		225,344				
7	AC Final Assy	ST-0601	74	54.41%	54.41%		
8		ST-0701	37	27.21%	81.61%		
9		AirFrame Std	13	9.56%	91.17%		
10		Airframe Extd	12	8.82%	100.00%		
11	Sum		136				
12	Albany	Gamer	10,175	100.00%	100.00%		
13	Sum		10,175				
14	AmeriComp	Workstation	84,000	64.24%	64.24%		
15		Server	36,400	27.84%	92.08%		

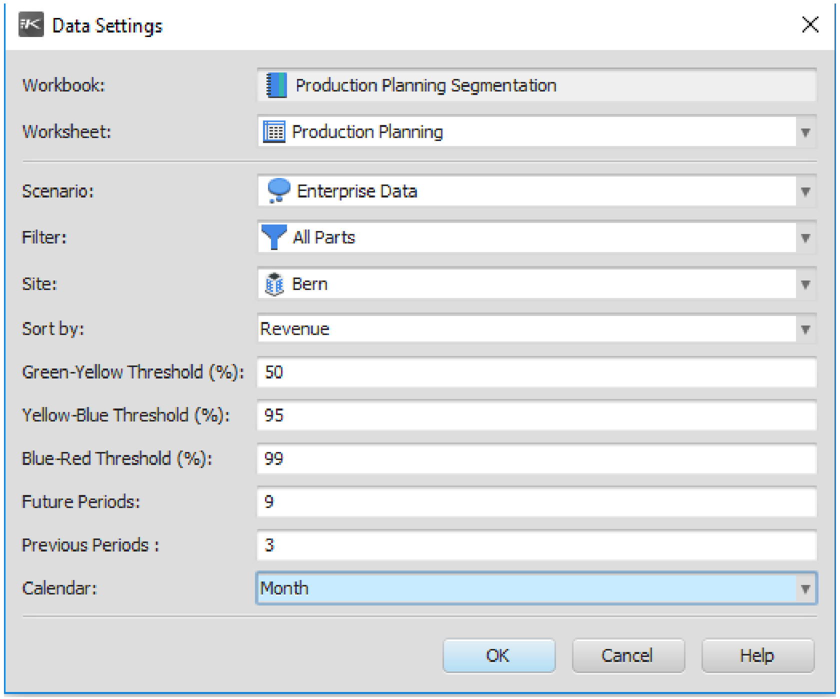
Sorted by Revenue – single site selected

Start | Production Planning Segmentation

Demo | All Parts | Bern | Reset Thresholds | Sort by: Revenue

	Part	Demand						Segment	
		Units	%	Cumulative	Revenue	%	Cumulative	Current	New
1	CRUISER	12,190	3.74%	3.74%	$35,225,998	37.82%	37.82%	Green	
2	RACER	13,275	4.07%	7.81%	$26,729,250	28.70%	66.52%	Green	
3	MOUNTAIN	10,209	3.13%	10.94%	$21,540,105	23.13%	89.65%	Yellow	
4	WHEEL-03	27,630	8.48%	19.42%	$2,644,306	2.84%	92.49%	Yellow	
5	STEER-01	19,100	5.86%	25.28%	$1,222,810	1.31%	93.80%	Yellow	
6	C-FRAME-STD	10,445	3.20%	28.48%	$906,827	0.97%	94.78%	Yellow	
7	M-FRAME-STD	8,655	2.66%	31.14%	$751,254	0.81%	95.58%	Yellow	
8	WHEEL-01	20,890	6.41%	37.55%	$716,106	0.77%	96.35%	Blue	
9	SEAT-01	16,725	5.13%	42.68%	$572,556	0.61%	96.97%	Blue	
10	R-FRAME-STD	6,280	1.93%	44.61%	$543,264	0.58%	97.55%	Blue	
11	BRAKE-01	25,380	7.79%	52.39%	$520,434	0.56%	98.11%	Blue	
12	DR-0701	50,760	15.57%	67.97%	$467,685	0.50%	98.61%	Blue	
13	FR-0201	8,062	2.47%	70.44%	$458,439	0.49%	99.10%	Blue	
14	DR-0601	25,380	7.79%	78.23%	$233,843	0.25%	99.35%	Red	
15	DR-0801	19,100	5.86%	84.09%	$219,407	0.24%	99.59%	Red	
16	DR-0802	6,280	1.93%	86.02%	$172,050	0.18%	99.78%	Red	
17	DR-0822	17,582	5.39%	91.41%	$79,108	0.08%	99.86%	Red	
18	WH-0501	13,400	4.11%	95.52%	$60,080	0.06%	99.92%	Red	
19	ST-0601	6,280	1.93%	97.45%	$28,274	0.03%	99.96%	Red	
20	DR-0823	2,020	0.62%	98.07%	$23,028	0.02%	99.98%	Red	
21	ST-0701	6,280	1.93%	100.00%	$14,179	0.02%	100.00%	Red	
22	**Sum (Site)**	**325,923**			**$93,129,002**				

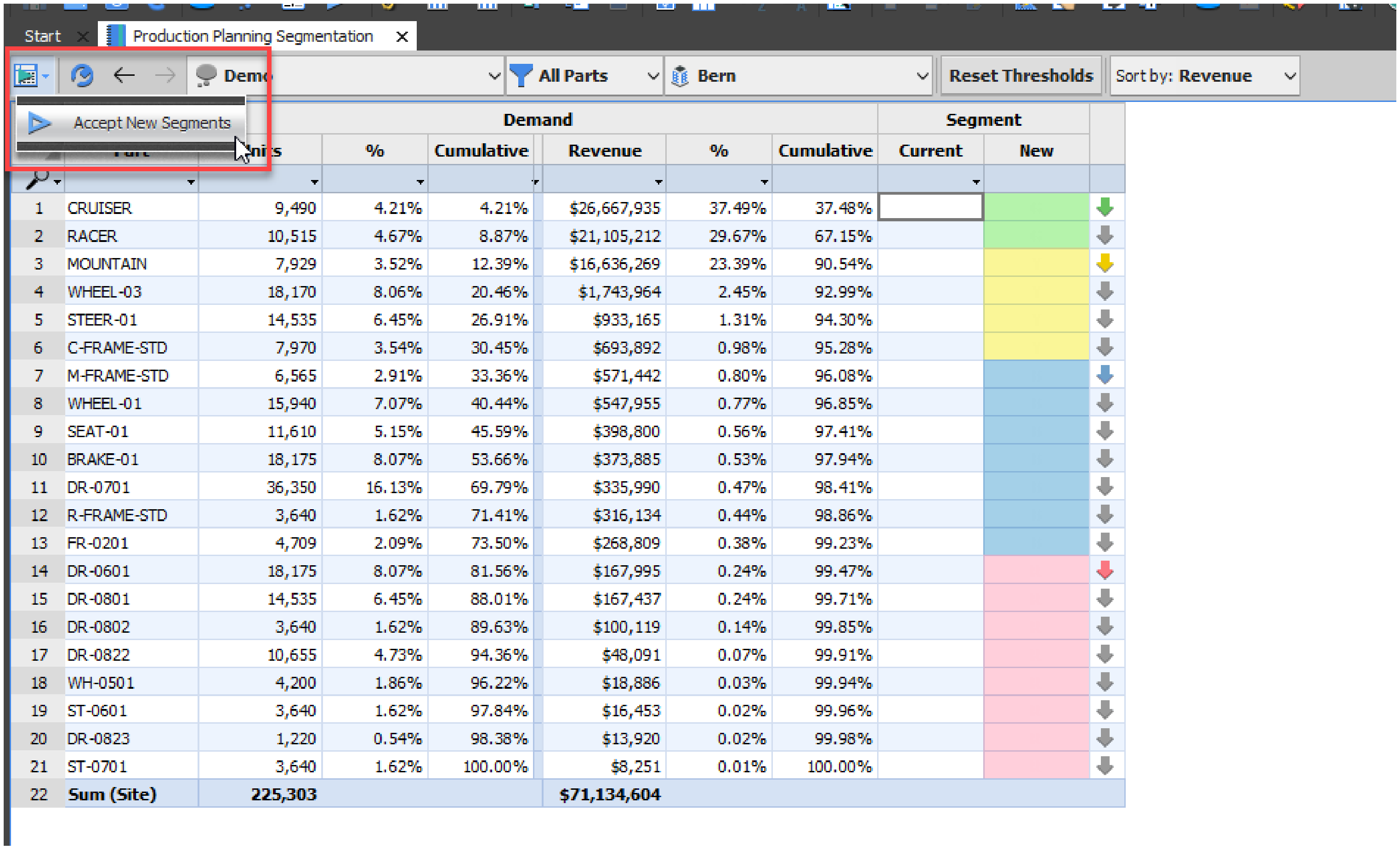
Out of the box this resource allows to tag parts with a segment following a strategy inspired on the Glenday Sieve. Out of the box, there are 4 segments pre-defined: Green, Yellow, Blue, and Red. Thresholds are based on the cumulative Demand (units or revenue) over a period defined by Future and Previous periods variables.



Out of the box defined as:

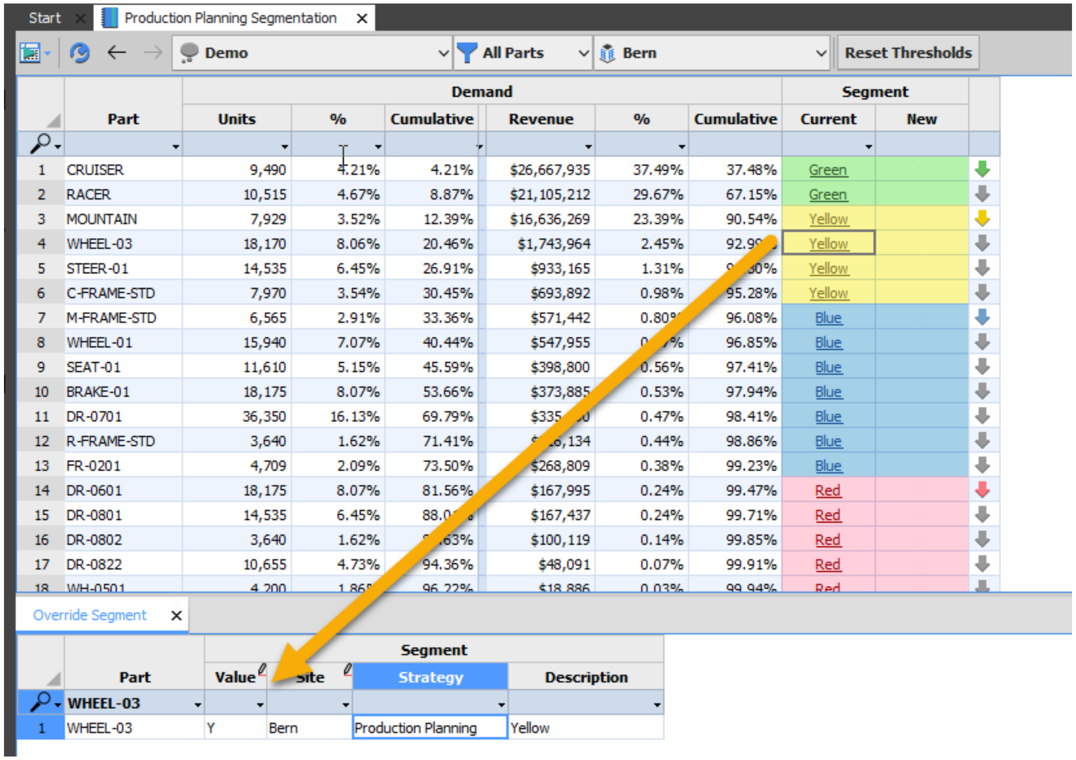
Segment	Threshold
Green	<50% of total cumulative demand
Yellow	>=50%, <95%
Blue	>=95%, <99%
Red	>=99%

These thresholds can be changed from either the Data Settings dialog, or by clicking on the arrows located at the rightmost column (when a single site is selected).



	Part	Units	%	Cumulative	Revenue	%	Cumulative	Current	New
		Demand						Segment	
1	CRUISER	9,490	4.21%	4.21%	$26,667,935	37.49%	37.48%		
2	RACER	10,515	4.67%	8.87%	$21,105,212	29.67%	67.15%		
3	MOUNTAIN	7,929	3.52%	12.39%	$16,636,269	23.39%	90.54%		
4	WHEEL-03	18,170	8.06%	20.46%	$1,743,964	2.45%	92.99%		
5	STEER-01	14,535	6.45%	26.91%	$933,165	1.31%	94.30%		
6	C-FRAME-STD	7,970	3.54%	30.45%	$693,892	0.98%	95.28%		
7	M-FRAME-STD	6,565	2.91%	33.36%	$571,442	0.80%	96.08%		
8	WHEEL-01	15,940	7.07%	40.44%	$547,955	0.77%	96.85%		
9	SEAT-01	11,610	5.15%	45.59%	$398,800	0.56%	97.41%		
10	BRAKE-01	18,175	8.07%	53.66%	$373,885	0.53%	97.94%		
11	DR-0701	36,350	16.13%	69.79%	$335,990	0.47%	98.41%		
12	R-FRAME-STD	3,640	1.62%	71.41%	$316,134	0.44%	98.86%		
13	FR-0201	4,709	2.09%	73.50%	$268,809	0.38%	99.23%		
14	DR-0601	18,175	8.07%	81.56%	$167,995	0.24%	99.47%		
15	DR-0801	14,535	6.45%	88.01%	$167,437	0.24%	99.71%		
16	DR-0802	3,640	1.62%	89.63%	$100,119	0.14%	99.85%		
17	DR-0822	10,655	4.73%	94.36%	$48,091	0.07%	99.91%		
18	WH-0501	4,200	1.86%	96.22%	$18,886	0.03%	99.94%		
19	ST-0601	3,640	1.62%	97.84%	$16,453	0.02%	99.96%		
20	DR-0823	1,220	0.54%	98.38%	$13,920	0.02%	99.98%		
21	ST-0701	3,640	1.62%	100.00%	$8,251	0.01%	100.00%		
22	**Sum (Site)**	**225,303**			**$71,134,604**				

The "New" columns would show the recommended segment based on the threshold values. In order to accept these recommendations, a command needs to run. The "Accept New Segments" will write the segment value to the PartSolutions table.



Segments can be overridden by drilling from the Current Column and editing the Value field in the Override Segment worksheet.

Note: Segments currently do not affect the production planning analytics directly, therefore changes made to a part's segment will not immediately impact the production plan. Segments are to be used in conjunction with

4.4. Roles and Responsibilities

- Resources are shared with the Production Planner group.
- Wheel responsibility follows constraint responsibility, no new responsibilities definition created.